Factoring trinomials

1

a No

b Yes

c Yes

d No

2

a $(5+v)(a+b)$

b $(y-z)(x-w)$

c $(2y+3w)(8x-z)$

d $\left(5x^2+4y\right)(8z-1)$

e $2(y-4)(4y-5)$

f $(4c-d)(c+9d+5)$

3

a $(x+5)(x+8)$

b $(x-5)(x+10)$

4

a $(x+2)(y-9)$

b $(8-x)(7-y)$

c $(x+z)(12y+w)$

d $(x-20y)(2+z)$

e $(5+x)(10+y)$

f $(12y+w)(x+z)$

5 $(x+4)(4x+1)$

6 To factor the quadratic $x^2+11x+24$, we need to find two numbers whose product is 24 and whose sum is 11.

7

a $a=7, b=8$

b $a=-10, b=-5$

8

a $a=-2, b=-9$

b $a=8, b=-14$

9

a $m+10$

b $y+5$

10 $(m+5)(m+9)$

11

a $(x+5)(x+4)$

b $(3x+1)(x+6)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12

a $(x+1)(x+5)$ b $(x+4)(x+10)$ c $(x-1)(x-4)$ d $(x-3)(x-10)$

e $(x-8)(x-7)$ f $(x+12)(x-11)$ g $(x+15)(x-6)$ h $(x+2)(x-4)$

i $(x+7)(x-10)$ j $(x+2)(x+3)$ k $(x-11)(x-4)$ l $-(x+1)(x+12)$

m $-(x-9)(x-1)$

13

a $4(x+6)(x+8)$ b $-5(x-4)(x+2)$ c $3(x-9)(x+2)$ d $2x(x+5)(x+3)$

14 $-10(t+2)(t-7)$

15

a $-16(t+1)(t-2)$

b 0

c 0

d $-16t^2+16t+32$ and $-16(t+1)(t-2)$ evaluated at $t=2$ are both equal to 0. This means that 2 seconds after jumping off the board, Tara has hit the water.

16 $(4+x)(3-x)$ square units

17

a $3x^2+10x-8=(3x-2)(x+4)$ b $12x^2+25x+7=(4x+7)(3x+1)$

c $15x^2-13x-6=(5x-6)(3x+1)$

18

a $(7x-5)(x-10)$ b $(6x+7)(x-1)$

c $(9x-10)(x-1)$ d $(7x+8)(x-6)$

e $(3x+2)(2x+3)$ f $-(4x-3)(3x+4)$

g $(9-8x)(10-x)$ or $(8x-9)(x-10)$ h $-(7p-2)(7p+4)$

19 $2x+5, 3x+4$

20

a $10y^2-112y+22$ b $2(5y-1)(y-11)$

21 $9, -9, 15, -15$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

22 $3, 4$

Additional questions

23

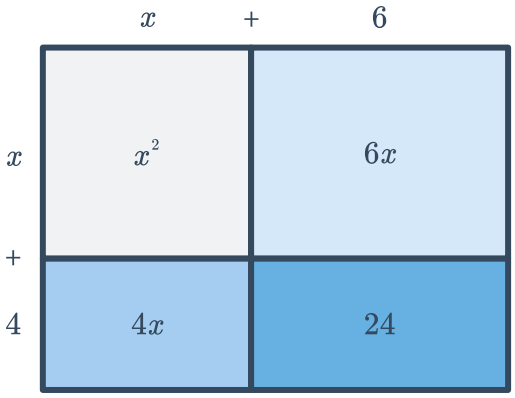
a $(x-6)(x+7)$

b x^2+x-42

c The sum of 7 and -6 is 1. The product of 7 and -6 is -42.

24

a



b $(x+6)(x+4)$

25

a Exactly one of p or q must be negative.

b Both p and q must be positive.

c Both p and q must be negative.

26

a $(x+9)(x+8)=x^2+17x+72$

b $(x+2)(x+1)=x^2+3x+2$

c $(x+1)(x-9)=x^2-8x-9$

27

a To maximize the length of the entire field you would increase the length by 36 and the width by 1

b If both portions are extended by 6 then the areas in the bottom left and top right will be the same, which would minimize the difference in area.

28

a 15

b The same

c -16

d Opposites

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

29

a -45 **b** Opposites **c** -4 **d** Opposites

30 Although this expression appears to have a leading coefficient of 3, the 3 is a common factor that can be factored out of the expression which leaves a trinomial where $a = 1$. The fully factored form is $3\left(x^2 + 5x - 21\right) = 3\left(x + 7\right)\left(x - 5\right)$.

31

a Answers may vary.
$4x^2 + 12x + 5 = (2x + 1)(2x + 5)$ is one possible answer.

32

a Each polynomial has a leading coefficient $a \neq 1$ and none of the polynomials can be factored.

b Find the product $a \times c$, then look for factors of ac that can add or subtract to be equal to the value of b. If no such factors exist, the polynomial is not factorable.

33

a She would use x to represent the width of her border if she did not know yet what size border she wanted.

b The expression for area factors as $(2x + 5)\left(2x + 7\right))$ so the dimensions of the photograph are $2x + 5$ and $2x + 7$

c If the border is a width of x you would subtract $2x$ from each dimension since there is a border on both sides. That leaves the dimensions as 3 and $2x - 1$ so the area of the photograph is $6x - 3$ square units.

34 Answers may vary.
In both cases we are looking for a pair of binomials where the leading coefficients of the factors will multiply to be a and the constant terms will multiply to be c. The difference is that in a polynomial where $a \neq 1$ the factors of the constant c won't add up to the coefficient of the linear term, b.

35

a The largest possible value for b is 71 and the factors are $(12x - 1)\left(x + 6\right)$

b The smallest possible value for b is 1 and the factors are $(4x + 3)\left(3x - 2\right)$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647